

PYTHON - SE - BACKEND

SOFTWARE ENGINEERING – MODULE 4

Database Management – SQL & PL/SQL

SESSION / ASSIGNMENT – 5

TASKS :

1. Create two tables in SQL: influencers (with influencer_id as PRIMARY KEY and name) and posts (with post_id as PRIMARY KEY, influencer_id as FOREIGN KEY, and caption). Insert at least 3 influencers and 2 posts for each influencer.

→ CODE:

```
CREATE TABLE influencers (  
    influencer_id INT PRIMARY KEY,  
    name VARCHAR(100)  
);
```

```
CREATE TABLE posts (  
    post_id INT PRIMARY KEY,  
    influencer_id INT,  
    caption VARCHAR(255),  
    FOREIGN KEY (influencer_id) REFERENCES influencers(influencer_id)  
);
```

```
INSERT INTO influencers (influencer_id, name)  
VALUES  
(1, 'Dhruv Mistry'),  
(2, 'Raj Patel'),  
(3, 'Isha Sharma');
```

```
INSERT INTO posts (post_id, influencer_id, caption)  
VALUES  
(101, 1, 'Exploring new places!'),
```

(102, 1, 'My latest adventure!'),
(103, 2, 'Fitness motivation today!'),
(104, 2, 'New day, new goals!'),
(105, 3, 'Creating something amazing!'),
(106, 3, 'Weekend vibes!');

SELECT * FROM influencers;
SELECT * FROM posts;

OUTPUT:

Show query box

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0069 seconds.)

CREATE TABLE influencers (influencer_id INT PRIMARY KEY, name VARCHAR(100));

Edit inline Edit Create PHP code

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0059 seconds.)

CREATE TABLE posts (post_id INT PRIMARY KEY, influencer_id INT, caption VARCHAR(255), FOREIGN KEY (influencer_id) REFERENCES influencers(influencer_id));

Edit inline Edit Create PHP code

3 rows inserted. (Query took 0.0009 seconds.)

INSERT INTO influencers (influencer_id, name) VALUES (1, 'Dhruv Mistry'), (2, 'Raj Patel'), (3, 'Isha Sharma');

Edit inline Edit Create PHP code

6 rows inserted. (Query took 0.0010 seconds.)

INSERT INTO posts (post_id, influencer_id, caption) VALUES (101, 1, 'Exploring new places!'), (102, 1, 'My latest adventure!'), (103, 2, 'Fitness motivation today!'), (104, 2, 'New day, new goals!'), (105, 3, 'Creating something amazing!'), (106, 3, 'Weekend vibes!');

Edit inline Edit Create PHP code

Showing rows 0 - 2 (3 total, Query took 0.0002 seconds.)

SELECT * FROM influencers;

Profiling Edit inline Edit Explain SQL Create PHP code Refresh

Showing rows 0 - 2 (3 total, Query took 0.0002 seconds.)

SELECT * FROM influencers;

Profiling Edit inline Edit Explain SQL Create PHP code Refresh

Show all | Number of rows: 25 Filter rows: Search this table Sort by key: N

Extra options

← T →					influencer_id	name
☐	 Edit	 Copy	 Delete		1	Dhruv Mistry
☐	 Edit	 Copy	 Delete		2	Raj Patel
☐	 Edit	 Copy	 Delete		3	Isha Sharma

✓ Showing rows 0 - 5 (6 total, Query took 0.0003 seconds.)

```
SELECT * FROM posts;
```

☐ Profiling [[Edit inline](#)] [[Edit](#)] [[Explain SQL](#)] [[Create PHP code](#)] [[Refresh](#)]

☐ Show all | Number of rows: 25 ▼ Filter rows: Sort by key

Extra options

			post_id	influencer_id	caption
☐	Edit	Copy	Delete	101	1 Exploring new places!
☐	Edit	Copy	Delete	102	1 My latest adventure!
☐	Edit	Copy	Delete	103	2 Fitness motivation today!
☐	Edit	Copy	Delete	104	2 New day, new goals!
☐	Edit	Copy	Delete	105	3 Creating something amazing!
☐	Edit	Copy	Delete	106	3 Weekend vibes!

2. Write an SQL query using INNER JOIN to display each post's caption along with the name of the influencer who posted it, based on the tables you created.

➔ CODE:

```
SELECT posts.caption, influencers.name
FROM posts
INNER JOIN influencers
ON posts.influencer_id = influencers.influencer_id;
```

OUTPUT:

✓ Showing rows 0 - 5 (6 total, Query took 0.0038 seconds.)

```
SELECT posts.caption, influencers.name FROM posts INNER JOIN influencers ON posts.influencer_id = influencers.influencer_id;
```

☐ Profiling [[Edit inline](#)] [[Edit](#)] [[Explain SQL](#)] [[Create PHP code](#)] [[Refresh](#)]

☐ Show all | Number of rows: 25 ▼ Filter rows: Sort by key: None ▼

Extra options

caption	name
Exploring new places!	Dhruv Mistry
My latest adventure!	Dhruv Mistry
Fitness motivation today!	Raj Patel
New day, new goals!	Raj Patel
Creating something amazing!	Isha Sharma
Weekend vibes!	Isha Sharma

3. Write an SQL query using LEFT JOIN to show all influencers and their posts, including influencers who haven't posted anything yet. If an influencer has no posts, display 'No Posts' in the caption column.

Hint: Use IFNULL or COALESCE to handle NULL values in the result.

➔ CODE:

```
SELECT influencers.name,  
COALESCE(posts.caption, 'No Posts') AS caption  
FROM influencers  
LEFT JOIN posts  
ON influencers.influencer_id = posts.influencer_id;
```

OUTPUT:

Showing rows 0 - 5 (6 total, Query took 0.0014 seconds.)

```
SELECT influencers.name, COALESCE(posts.caption, 'No Posts') AS caption FROM influencers LEFT JOIN posts ON influencers.influencer_id = posts.influencer_id;
```

☐ Profiling [\[Edit inline \]](#) [\[Edit \]](#) [\[Explain SQL \]](#) [\[Create PHP code \]](#) [\[Refresh \]](#)

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

name	caption
Dhruv Mistry	Exploring new places!
Dhruv Mistry	My latest adventure!
Raj Patel	Fitness motivation today!
Raj Patel	New day, new goals!
Isha Sharma	Creating something amazing!
Isha Sharma	Weekend vibes!

4. Write an SQL query using RIGHT JOIN to list all posts and the corresponding influencer's name, ensuring that even posts without a matching influencer_id (if any) are shown.

→ CODE:

```
SELECT posts.post_id, posts.caption, influencers.name
FROM influencers
RIGHT JOIN posts
ON influencers.influencer_id = posts.influencer_id;
```

OUTPUT:

✓ Showing rows 0 - 5 (6 total, Query took 0.0008 seconds.)

```
SELECT posts.post_id, posts.caption, influencers.name FROM influencers RIGHT JOIN posts ON influencers.influencer_id = posts.influencer_id;
```

☐ Profiling [[Edit inline](#)] [[Edit](#)] [[Explain SQL](#)] [[Create PHP code](#)] [[Refresh](#)]

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

post_id	caption	name
101	Exploring new places!	Dhruv Mistry
102	My latest adventure!	Dhruv Mistry
103	Fitness motivation today!	Raj Patel
104	New day, new goals!	Raj Patel
105	Creating something amazing!	Isha Sharma
106	Weekend vibes!	Isha Sharma